Lab 11: Genomics & Biotechnology

BIOL-1

Name: ______________________________ **Date:** ______________________________

Overview

In this lab, you will learn the core tools of modern biotechnology: cutting DNA, copying DNA, sorting DNA, and using DNA to identify people and treat diseases.

Materials

- This worksheet
- Ruler (optional, for gel diagram)

Part 1: Restriction Enzymes and Recombinant DNA

Restriction enzymes are "molecular scissors" that cut DNA at specific sequences. DNA ligase is "molecular glue" that joins DNA pieces together.

1. What does a restriction enzyme do?

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2. What are "sticky ends"? Why are they useful?

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3. What does DNA ligase do?

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4. What is recombinant DNA?

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5. Put these steps in order to create a GMO bacterium that makes human insulin:

Transform the plasmid into bacteria
Cut the plasmid and human DNA with the same restriction enzyme

Use ligase to seal the human gene into the plasmid
Select for bacteria that took up the plasmid
____ Mix the cut plasmid and human gene together

Part 2: PCR (Polymerase Chain Reaction)

PCR is like a "molecular photocopier" — it makes millions of copies of a specific piece of DNA.

6. Fill in the table:

PCR Step	Temperature	What Happens
Denaturation	~95°C	
Annealing	~55°C	
Extension	~72°C	

7. Why does PCR use Taq polymerase instead of normal DNA polymerase?

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8. Starting with 1 DNA molecule, how many copies do you have after:

- 1 cycle: ____
- 2 cycles: ____
- 3 cycles: ____
- 10 cycles: ____

Part 3: Gel Electrophoresis

Gel electrophoresis separates DNA pieces by size using electricity.

9. DNA is negatively charged. Does it move toward the positive (+) or negative (–) end?

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10. Which travels farther through the gel: small fragments or large fragments?

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11. You run three DNA samples on a gel. Sample A has one band near the top. Sample B has one band near the bottom. Which sample has larger DNA fragments?

Part 4: DNA Fingerprinting

12. What are STRs (Short Tandem Repeats)?

13. A child has STR bands at positions 1, 2, 3, and 4. The mother has bands at 1 and 3. Which bands must have come from the biological father?

14. A crime scene sample matches Suspect B's STR profile exactly but does not match Suspects A or C. What can you conclude?

Part 5: Gene Therapy and CRISPR

15. What is gene therapy?

16. What is the difference between ex vivo and in vivo gene therapy?

- Ex vivo: _______________
- In vivo: _______________

17. What is CRISPR-Cas9?

18. What is the difference between somatic gene therapy and germline modification?

19. Why is germline modification considered more controversial than somatic gene therapy?

20. The genetic code is nearly universal. Why does this make biotechnology possible? (Hint: Think about bacteria and human genes.)